

B.E. II Yr Examination Dec 2023
II yr (Electronics & Instrumentation Engineering (Regular))
Sub: Data Structure (3EIRC3)

Time: 3 hrs

Marks: 60

Note: All questions carry equal marks. Attempt any two part from each question.

Q.1

A) Explain the criteria of selection of Data Structure and its importance with a daily life example. 6

B) Write a program for Insertion Sort technique. 6

C) Write a program for Linked List Implementation with Insertion, Deletion and Traversing operation. 6

Q.2

A) Evaluate the given expression using Stack into postfix form showing Stack status after every step in tabular form: $[A + (B * C - (D / E - F) * G) * H]$ 6

B) Suppose the array a contain 12 elements as given below. Apply Merge Sort algorithm to sort the elements. 85, 76, 46, 92, 30, 41, 12, 19, 93, 3, 50, 11 6

C) Write a program for static implementation of Stack with PUSH, POP & Traverse operation. 6

Q.3

A) Explain the Mid Square Hash function, Division Hash function, Multiplication Hash function and Uniform Hash function in detail. 6

B) Discuss Single Ended Queue, Double Ended Queue and Circular Queue with their applications. 6

C) Write a program for implementation of Queue for different operation. 6

Q.4

A). Assume the list of following data elements and make the Height Balanced Tree. 6
[B, F, I, N, L, P, J, A, H, D]

B) Discuss Strictly Binary Tree and Complete Binary Tree in detail with an example. 6

C). Write a program for Preorder, Postorder and Inorder traversal technique of Binary Tree. 6

Q.5

A). Write a code for Dijkstra's algorithm using adjacent list for a Graph. 6

B) Explain the Shortest Path Problem with an example. 6

C) Explain the Adjacent Matrix and Adjacent list representation of Graph with an example. 6